One-loop corrections for the bad Boussinesq soliton and breather

Project *quantum_boussinesq_solitons* (Phase 2, next step 2)

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Abstract

We ask whether Dashen–Hasslacher–Neveu (DHN) one-loop corrections can fix the $O(\hbar^0)$ ambiguities of the candidate S-matrix in `smatrix_ansatz.pdf`.

Step 0: well-posedness. The one-loop problem is ill-posed on the full phase space. For the unstable phonons ($|q| > 1$) the soliton changes the amplitude of the mode, not only its phase, so no phase shift exists. The problem is well posed and finite on the Charlier–Lenells invariant class, where only stable phonons $|q| < 1$ are kept. The PI chose this "scheme A".

Results in scheme A.

- The phonon phase shifts are $\theta(q) = \arg A_{12}$, verified against the linearised equation.
- The one-loop soliton energy $E_1(\beta)$ is finite and unambiguous at fixed β. The zero-point momentum needs a prescription at the band edges.
- The breather's one-loop energy and momentum are the holomorphic continuations of the constituents' values.
- Consequently the string condition of the ansatz needs *no* $O(\hbar^0)$ shift and no $\hbar$ renormalisation at this order, once rapidities label the one-loop dispersion. The Maslov index of the breather orbit is assumed, not derived.

Main caveat. The short-wave part of the stable band, $1/\sqrt{2} < |q| < 1$, has reversed group velocity. Its phonon at $q^* = -\sqrt{3}/2$ co-moves with a slow soliton and is captured, so $E_1 - VP_1 \to -\frac{3\sqrt{3}}{8}\hbar$ as $\beta \to 0$. The quantum Toda chain has no such mode. Scheme A therefore differs from the Toda continuum limit at one loop even for small amplitudes.

We also correct an error in `smatrix_ansatz.pdf`: the breathers with $\pi/3 < b < \pi/2$ move to the left, not the right.

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1 Step 0: is the one-loop problem well posed?

Phonons as continued solitons. A soliton has $\tau = 1 + e^{Kx-\Omega t}$, $\Omega^2 = K^2 + K^4$. At $K = iq$ this becomes a phonon $e^{iqx-i\omega t}$ with $\omega^2 = q^2 - q^4$. We take $\beta = i \arcsin q$ for $|q| < 1$ and $\beta = i\pi/2 + \operatorname{arccosh} q$ for $|q| > 1$. Hirota's two-soliton τ, linearised in the phonon amplitude, gives an exact solution of the linearised equation around the soliton; the phonon emerges multiplied by $A(\text{soliton}, \text{phonon})$.

Proposition 1.1 (**(c)**, `scripts/oneloop_step0_wellposedness.py`)**.** *For solitons $\beta_s = 0.3, 0.7, 1.4$, both phonon branches:*

1. *for $|q| < 1$, $|A| = 1$ exactly, so $\theta(q) = \arg A$ is a real phase shift; the spectral parameter ℓ of the phonon is real, in $[-1, 1]$;*
2. *for $|q| > 1$, $|A| \neq 1$ (e.g. 1.84 at $q = 1.3$, $\beta_s = 0.7$), so the soliton changes the* amplitude *of the growing mode and no phase shift exists; ℓ is purely imaginary there.*

So the DHN sum over phase shifts is undefined on the full phase space: the full problem is *not* well posed. The unstable modes carry imaginary ℓ. We identify this, **(d)**, with the part $k \in [0, i]$ of the Charlier–Lenells spectral contour on which they impose $r_1 = 0$ to obtain global well-posedness **(a)** [2, 3]. The class $r_1|_{[0,i]} = 0$ is invariant under the flow. Its tangent space at a soliton is spanned by the stable phonons $|q| < 1$ and the zero modes.

Decision (PI). Three ways out were considered:

- **(A)** quantise the reduced (Charlier–Lenells) theory, keeping phonons $|q| < 1$ only;
- **(B)** use the Toda lattice as the definition, which is valid only for $\beta \to 0$;
- **(C)** analytic continuation from the good equation.

The PI chose (A). In (A) everything is finite without a free cutoff:

- $\omega(\pm 1) = 0$, so the band edges cost no zero-point energy.
- $\langle \phi_x^2 \rangle = \hbar/(2\pi)$ in the reduced vacuum.
- The only scheme freedom is normal ordering of ϕ_x^3. This adds $3\langle \phi_x^2 \rangle M = (3\hbar/2\pi)M$. Since $\int \phi_x = M$ is a boundary term, it does not change the equations of motion.

(c)

2 Phase shifts and the one-loop soliton energy

Proposition 2.1 (**(c)**, `scripts/oneloop_phase_shifts.py`)**.** *Let $\tau = 1 + e^{\theta_1} + \varepsilon(e^{\theta_2} + Ae^{\theta_1+\theta_2})$ and $\delta v = \partial_\varepsilon 2(\log \tau)_{xx}|_{\varepsilon=0}$. Then δv solves $\delta v_{tt} = \delta v_{xx} + \delta v_{xxxx} + 6(v_0 \delta v)_{xx}$; the residual is $< 10^{-29}$ at test points. Its asymptotic ratio $\delta v(+\infty)/\delta v(-\infty)$ equals A to 10^{-16}. The physical mode with $\omega > 0$ requires $\epsilon = \operatorname{sgn} q$ in the continuation: one oscillator per $q \in (-1, 1)$.*

The Levinson-type count $(2\pi)^{-1} \int_{-1}^{1} \theta' \, dq$ is *not* an integer: for example -0.63 at $\beta = 0.2$, -1.96 at $\beta = 0.7$ and $+0.57$ at $\beta = 1.5$. This is because θ does not vanish at the band edges. Edge modes have $\omega = 0$, so the lab-frame energy is unaffected; momentum is affected (see below).

Energy. Box quantisation $qL+\theta(q)=2\pi n$ gives, at fixed classical β **(b)** (DHN method) / **(c)** (here):

$$E_1(\beta)=\frac{\hbar}{4\pi}\int_{-1}^{1}\omega(q)\,\theta'(q)\,dq\ \ (+cM),\qquad \omega=|q|\sqrt{1-q^2}. \tag{1}$$

This is unambiguous, because $\omega=0$ at $q=0,\pm1$ where θ may jump by 2π.

β	0.01	0.03	0.1	0.3	0.7	1.1	1.5	2.0
$E_1/\hbar$	-0.2262	-0.2353	-0.2441	-0.2226	-0.1565	-0.1058	-0.0710	-0.0431

(c) (`scripts/oneloop_soliton_energy.py`; scheme term $c=0$. Normal ordering adds $(3\hbar/\pi)\sinh\beta$.)

Proposition 2.2 (**(c)**, `scripts/oneloop_soliton_energy.py`)**.** *As $\beta\to0$, $E_1\to-\frac{1}{2}\hbar\,\omega(q^*)=-\frac{\sqrt{3}}{8}\hbar$, with $q^*=-\sqrt{3}/2$. The mechanism is as follows.*

- *On $1/\sqrt{2}<|q|<1$ the band $\omega=|q|\sqrt{1-q^2}$* decreases*, so the group velocity is reversed and takes every value in $(0,\infty)$.*
- *At $q^*(V)$ a phonon co-moves with the soliton ($V=\cosh\beta\to1$ gives $q^*=-\sqrt{3}/2$).*
- *Its phase shift winds by -2π in a window of width $O(\beta)$: $-0.996\times2\pi$ within $|q-q^*|<0.08$ at $\beta=10^{-3}$. One mode is captured.*
- *The approach to the limit is slow ($E_1+\omega(q^*)/2=-0.0097\hbar$ at $\beta=0.01$).*

Momentum. The zero-point momentum $P_1=\frac{\hbar}{4\pi}\int q\theta'\,dq$ has edge terms $q\theta|_{\pm1}\neq0$, which oscillate with the box length. We average them out; this is a prescription **(d)**. With it, at fixed physical momentum

$$E(P)=E_{\rm cl}(P)+[E_1-VP_1](\beta(P)),\qquad [E_1-VP_1]/\hbar\to-\tfrac{1}{2}\big(\omega(q^*)-q^*\big)=-\tfrac{3\sqrt{3}}{8}\approx-0.650,$$

numerically -0.640 at $\beta=0.01$, -0.562 at 0.1, -0.427 at 0.3 and -0.260 at 0.7. **(c)**

3 Breathers

Kinematics (correction). For $\beta_{1,2}=a\mp ib$, the envelope moves with

$$V=\frac{\operatorname{Re}w}{\operatorname{Re}k}=\frac{\cosh a\cos 2b}{\cos b}.$$

On $\pi/3<b<\pi/2$, $V<-\cosh a$: every breather is *supersonic and left-moving*, and none is at rest. The range $\pi/2<b<2\pi/3$ gives their parity images. **(c)** (`scripts/oneloop_breather.py`, checked against the centre of mass of v^2 over one period.) `smatrix_ansatz.pdf` had called them right-moving; its text and `smatrix_poles_bootstrap.py` have been corrected. The pole analysis there used only $\operatorname{Im}(P_1-P_2)$ and is unaffected.

Proposition 3.1 (**(c)**, `scripts/oneloop_breather.py`)**.** *1. $|A(\beta_1,\cdot)A(\beta_2,\cdot)|=1$ on the band, so the breather phase shift is $\theta_{\rm br}=\theta(\beta_1)+\theta(\beta_2)$, which is real.*

2. *Define $e(\beta)=\frac{1}{4\pi}\int\omega\,\partial_q(-i\log A)\,dq$ and $p(\beta)=\frac{1}{4\pi}\int q\,\partial_q(-i\log A)\,dq$. For real β these equal $E_1/\hbar$ and $P_1/\hbar$. Both satisfy Cauchy–Riemann along $\beta=a-it$, $0\le t\le b$, so no zero or pole of A crosses the band.*
3. *The breather's one-loop energy and momentum, computed directly from $\theta_{\rm br}$, equal $\hbar[e(\beta_1)+e(\beta_2)]$ and $\hbar[p(\beta_1)+p(\beta_2)]$.*

Semiclassical quantisation. The breather is a relative periodic orbit, with stability angles $T(\omega - Vq)$. Gutzwiller/DHN quantisation then gives **(b)** (method) / **(c)** (application):

$$E = E_{\rm cl} + \hbar \sum_i e(\beta_i), \qquad P = P_{\rm cl} + \hbar \sum_i p(\beta_i), \qquad \text{at } I_{\rm br}(\beta_1, \beta_2) = (N + \mu/4)\hbar.$$

We take $\mu = 0$, as in Bogdan–Charkina [1] and as DHN found for sine-Gordon. This is *not* derived here.

N	b_N	$E_{\rm cl}$	$\hbar \sum e$	$P_{\rm cl}$	$\hbar \sum p$
1	1.06092	0.87107	−0.01386	−0.50737	−0.00697
2	1.07467	0.97592	−0.01351	−0.38750	−0.00724
3	1.08846	1.07585	−0.01315	−0.26506	−0.00751

One-loop breather data at the ansatz poles, $\hbar = 0.1$, $a = 0.8$ (**(c)**, numerical; scheme $c = 0$). Direct breather sums agree with the continued constituent sums to 10^{-10}, e.g. $(E_1, P_1)/\hbar = (-0.10213, -0.09437)$ at $(a, b) = (0.8, 1.2)$.

4 Consequences for the S-matrix ansatz

Proposition 4.1 (**(c)** given $\mu = 0$)**.** *Label one-soliton states by their classical β, with one-loop dispersion $E_{\rm cl} + \hbar e$ and $P_{\rm cl} + \hbar p$. Then the poles of* `smatrix_ansatz.pdf` *at $I_{\rm br} = N\hbar$ carry exactly the one-loop breather energies and momenta. At this order:*

- *the string condition needs no shift, so $\Gamma(1 + i\eta)$ is confirmed rather than $\Gamma(\frac{1}{2} + i\eta)$, given $\mu = 0$;*
- *no renormalisation of $\hbar$ in η is needed (that would first appear at $O(\hbar^2)$ in the spectrum);*
- *the scheme term cM shifts soliton and breather energies consistently, since $M_{\rm br} = M_1 + M_2$.*

This holds because of two features: the breather phase shift is additive over constituents, and the continuation in β is holomorphic. Neither was guaranteed: a zero of A crossing the band would have produced an extra, CDD-like term.

What is not fixed.

- The Maslov index μ.
- The one-loop two-soliton phase. It requires the $O(\hbar)$ terms of Δx (two-phonon dressing), which were not computed. So CDD factors with $\log f = O(\hbar^0)$ that leave the bound-state energies unchanged remain undetermined.
- The edge prescription for P_1.

5 Cross-check with the quantum Toda chain

`toda_route.pdf` gives the Toda one-loop dispersion $E_1 = +0.0301$ and $+0.0203$ at $P = 0.6$ and 0.4 (Toda units). It is positive and decreases towards small P. In scheme A, $E_1 - VP_1 \to -\frac{3\sqrt{3}}{8}\hbar \neq 0$ as $\beta \to 0$. The non-vanishing limit comes entirely from the captured co-moving phonon at $q^* = -\sqrt{3}/2$.

Toda has no such phonon. Its band $\omega = 2|\sin(q/2)|$ has group velocity $\leq 1 < V$, so no Toda phonon co-moves with a supersonic soliton. Scheme A and the Toda continuum limit therefore *disagree at one loop even in the small-amplitude corner.* **(c)** (mechanism) / **(d)** (that the Toda

$E_1 \to 0$ as $P \to 0$ is inferred from two data points, and a quantitative comparison in scaling units was not carried out).

This is the most important finding for the choice of quantum theory. The reversed-velocity part of the stable band, $1/\sqrt{2} < |q| < 1$, is where the Boussinesq dispersion $q^2 - q^4$ departs from its lattice origin. A scheme that also removes it, keeping $|q| < 1/\sqrt{2}$, would remove the resonance. It is not clear that such a restriction defines an invariant class. This is a decision for the PI. **(d)**

6 Summary

Question	Answer	Status
Well posed on full phase space?	No: unstable modes have $\lvert A\rvert \neq 1$, so no phase shift	**(c)**
Well posed on CL class (scheme A)?	Yes; finite, no free cutoff, scheme term cM	**(c)**/**(d)**
Phase shifts	$\theta = \arg A$, checked against the linearised equation	**(c)**
Soliton E_1	finite table; $\to -\frac{\sqrt{3}}{8}\hbar$ as $\beta \to 0$ (resonance)	**(c)**
Soliton P_1	needs an edge prescription	**(d)**
Breather one-loop	sum of holomorphically continued constituent values	**(c)**
Ansatz pole condition	no $O(\hbar^0)$ shift, no $\hbar$ renormalisation (given $\mu = 0$)	**(c)**/**(d)**
CDD factors	not fixed (needs the one-loop two-soliton phase)	open
Toda cross-check	qualitative disagreement at small amplitude	**(c)**/**(d)**

A Scripts

All are in `phase2/scripts`; each ends with `ALL CHECKS PASSED`.

- `oneloop_step0_wellposedness.py`: Prop. 1.1.
- `oneloop_phase_shifts.py`: linearised modes, phase shifts, Levinson-type counts.
- `oneloop_soliton_energy.py`: E_1 table, Prop. 2.2, $\langle \phi_x^2 \rangle$.
- `oneloop_breather.py`: breather kinematics, Prop. 3.1, table.

Failures along the way:

- Machine-float phonon wavenumbers broke a 25-digit dispersion check.
- Integration break-points outside the band produced complex values.
- The root-finder for q^* left the band.
- I first assumed a resonance window of width $O(\beta^2)$; it is $O(\beta)$.
- I first took $b = \pi/2$ as a "breather at rest". It is degenerate ($E_{\rm cl} = 0$), and no breather is at rest. This led to the velocity correction above.

References

[1] M. M. Bogdan, O. V. Charkina, arXiv:2502.14695, *Low Temp. Phys.* 51 (2025).

[2] C. Charlier, J. Lenells, On Boussinesq's equation for water waves, arXiv:2204.02365.

[3] C. Charlier, J. Lenells, Direct and inverse scattering for the Boussinesq equation with solitons, arXiv:2302.14593.